

CHL5233: Quarto in-class assignment

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In Class Assignment - September 28th

For this exercise, we will use the following two data sets.

1. `countries.txt`: A data set that contains a list of countries used in the paper “Implications of armed conflict for maternal and child health: A regression analysis of data from 181 countries for 2000–2019” and their three-digit ISO (International Organization for Standardization) code.
2. `regions.txt`: A data set that contains a list of all major countries in the world and their
 - a. ISO code (`country_code`)
 - b. region (`region`)
 - c. sub-region (`sub_region`)
 - d. numerical region code (`region_code`)
 - e. numerical sub-region code (`sub_region_code`)

The **final goal** is to display a **bar chart** that depicts how many countries are in each region.

Read in data and create an analytical data set

First, edit the code chunk below that reads in `countries.txt` to also display the number of rows and columns of the data.

```
countries <- read.table("../data/raw/countries.txt", header = TRUE)
nrow(countries)
```

```
[1] 186
```

```
ncol(countries)
```

```
[1] 2
```

There are 186 countries in this countries dataset

Second, read in `regions.txt` and also write a sentence that states how many countries are included in the data set.

```
regions <- read.table("../data/raw/regions.txt", header = TRUE)
nrow(regions)
```

```
[1] 249
```

```
ncol(regions)
```

```
[1] 6
```

There are 249 countries in this regions data set

Third, subset the `regions` data set to only include countries from the `countries` data set.

```
regions_subset <- regions |>
  filter(country_code %in% countries$ISO)
```

Tabulate countries by region and sub-region

Next, we want to find out how many countries are in a region. Use the `group_by()` and `summarise()` functions in the `dplyr` package to count the number of countries in each region. Then use the `kable()` function in the `knitr` package to create a simple table with the column headers, “Region” and “Count”.

```
regions_count <- regions_subset |>
  group_by(region) |>
  summarise(Count = n())

regions_count |>
  kable(col.names = c("Region", "Count"))
```

Region	Count
Africa	54
Americas	35
Asia	46
Europe	40

Region	Count
Oceania	11

Create a similar table that shows the number of countries in each **sub-region**.

```
sub_regions_count <- regions_subset |>
  group_by(sub_region) |>
  summarise(Count = n())

sub_regions_count |>
  kable(col.names = c("Sub-Region", "Count"))
```

Sub-Region	Count
Australia and New Zealand	2
Central Asia	5
Eastern Asia	5
Eastern Europe	10
Latin America and the Caribbean	33
Melanesia	4
Micronesia	3
Northern Africa	6
Northern America	2
Northern Europe	10
Polynesia	2
South-eastern Asia	11
Southern Asia	9
Southern Europe	13
Sub-Saharan Africa	48
Western Asia	16
Western Europe	7

Data visualization using ggplot

Here, we will use `geom_bar()` function within `ggplot()` to create a bar chart that displays the **number of countries** in each **sub-region**.

1. Color code each sub-region by **region**.
2. Create bars that run horizontally where sub-region names appear along the y-axis.
3. Order the bars such that the sub-region with the most number countries appear at the top and the sub-region with the least number of countries appear at the bottom.

```

ggplot(regions_subset,
       aes(x = reorder(sub_region, sub_region, FUN = length),
           fill = region)) +
geom_bar() +
coord_flip() +
labs(
  x = "Sub-Region",
  y = "Number of Countries",
  fill = "Region"
)

```

